



MOVING THE WRIST AND FINGERS
Slim flexor digitorum muscles on the inside of the forearm have long tendons that run through a "watch strap" of ligament fibers, known as the carpal tunnel in the wrist. These muscles flex the fingers, bending their tips toward the palm, as when playing the piano or making a fist.

Biceps brachii
bends elbow
and twists
wrist

Large deltoid muscle
lifts the entire limb

Vein takes stale blood
back towards the heart

Coracobrachialis
pulls the arm
toward the body

Jugular
vein

Part of serratus
anterior, which
helps arm to reach out

Nerves give instructions
to the muscles from the
brain (pp. 58-59)

Branches of
facial nerve

THE HEAD AND NECK

The head weighs about 8.8 lb (4 kg). To balance and move this heavy load on the topmost bones of the spine, numerous muscles attach the base of the skull to the vertebrae, shoulder blades, and other bones in the upper chest (pp. 14-15). About 30 muscles produce a galaxy of facial expressions, allowing people to communicate silently. Many of these muscles are not attached to bones, but to their fellow muscles. About 10 muscles meet near the corner of the mouth, anchored to a disk of fibrous tissue known as the modiolus. The eyelids and lips contain circlelets of muscle that close when they contract.

Trapezius prevents the
head from falling forward

Procerus
wrinkles bridge
of nose to frown

Levator labii superior
lifts and curls upper
lip to pucker

Orbicularis oris
closes lips and
purrs them

Modiolus
is at corner
of mouth

Mylohyoid forms
floor of mouth and
helps to swallow

Sternohyoid pulls
floor of mouth and
voice box down

Sternocleidomastoid
holds the head
steady and tilts it
up and sideways

Intercostal
muscles raise the
ribs in breathing

Ribs (p. 15)



SWAMMERDAM OF AMSTERDAM

Jan Swammerdam (1637-1680) qualified as a doctor but spent most of his working life peering down a microscope. He devised methods of making fresh frog muscles contract by pinching the nerves attached to them. By immersing the muscle in a water-filled device, he showed for the first time that when a muscle shortens, its total volume does not increase, as had long been believed, but stays the same.